

On Information Security in Bounded-Topology Discrete Communication Channels: Distinguishable Independent Coherent Communication Channels

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Abstract

We establish a rigorous mathematical framework for covert communication exploiting the discrete channel structure forced by bounded topological systems. Every bounded measure-preserving dynamical system necessarily exhibits a hierarchical partition with discrete coordinate structure (n, l, m, s) derived from $SO(3)$ geometry. This partition structure determines both the system’s capacity for independent communication channels and the mathematical invisibility of those channels to external observers measuring only aggregate observables. We prove three principal theorems: (I) the *Topological Channel Multiplicity Theorem*, stating that cycle rank C (or partition depth n) forces exactly $C+1$ (or $2n^2$) distinguishable independent communication channels at zero cost; (II) the *Capacity–Coherence Duality*, showing that maximum simultaneous message streams equals topology rank weighted by coherence window T_{deph}/T_L ; and (III) the *Observation Invisibility Theorem*, proving that external observers measuring only field projections cannot detect or decode hidden channels. We instantiate the framework in three physical regimes: (1) harmonic-edge coupling in molecular resonators, (2) partition-coordinate stacking in multi-layer fluids, and (3) spectral-transfer channels in optical systems. A unified communication protocol is constructed in which sender and receiver synchronise on a bounded topological system, encode independent message streams in orthogonal eigenmodes of the system’s topology, and achieve provably covert,

high-capacity information exchange. Numerical validation across synthetic molecular resonators ($C \in \{1, \dots, 10\}$), synthetic partition systems ($n \in \{4, \dots, 16\}$), and simulated optical stacks ($K \in \{2, \dots, 32\}$) confirms all predicted capacities and verifies that external observers remain unable to distinguish covert communication from background noise. The framework is information-theoretically secure in the sense that full knowledge of the aggregate observable field does not constrain the hidden message space. Applications include secure inter-agent coordination in distributed systems, covert navigation without explicit signals, and detection-resistant distributed computation.

Keywords: bounded systems, discrete topology, communication capacity, covert channels, harmonic structure, partition coordinates, information security, topological capacity.

Contents

1	Introduction	2
1.1	The Capacity-Topology Principle .	2
1.2	The Covertiness Mechanism	3
1.3	Three Instantiations	3
1.4	Security Model	3
1.5	Contributions	3
1.6	Roadmap	4
2	Mathematical Foundations of Bounded Systems	4
2.1	Bounded Measure-Preserving Dynamical Systems	4

2.2	Discrete Frequency Spectrum from Recurrence	4	8.4	Experiment 4: External-Observer Invisibility	15
2.3	Partition Coordinates from $SO(3)$ Geometry	5	9	Applications	17
2.4	Partition Potential and Dynamics .	5	9.1	Covert Inter-Agent Coordination in Distributed Networks	17
3	Topology and Channel Structure	6	9.2	Covert Navigation Without Explicit Signals	17
3.1	Cycle Rank of Bounded Systems .	6	9.3	Detection-Resistant Distributed Computation	17
3.2	Partition Depth and Cell Structure	6	9.4	Stealth Communication in Adversarial Environments	17
4	Channel Capacity and Information-Theoretic Bounds	6	10	Discussion	17
4.1	The Topological Channel Multiplicity Theorem	6	10.1	Limitations and Open Questions .	17
4.2	The Capacity-Coherence Duality .	8	10.2	Experimental Realisability	17
4.3	The Observation Invisibility Theorem	9	10.3	Relation to Existing Stealth Communication Methods	17
5	Physical Instantiations	9	11	Conclusion	18
5.1	Harmonic-Edge Channels in Molecular Resonators	9			
5.1.1	Harmonic Graph Structure	9			
5.1.2	Transfer Matrix Through Harmonic Couplings	9			
5.1.3	Message Encoding via Harmonic Multiplexing	10			
5.2	Partition-Coordinate Channels in Stacked Fluids	10			
5.2.1	Stacked-Fluid Optical System	10			
5.2.2	Message Encoding via Partition Coordinates	10			
5.3	Spectral-Transfer Channels in Frequency-Dispersive Systems . . .	10			
6	Unified Communication Protocol	11			
6.1	Protocol Architecture	11			
6.2	Error Correction and Robustness .	11			
7	Information-Theoretic Security Analysis	13			
7.1	Threat Model	13			
7.2	Information-Theoretic Bounds . . .	13			
7.3	Computational Security Against Side-Channel Analysis	13			
8	Numerical Validation	13			
8.1	Experiment 1: Harmonic-Graph Channel Capacity	13			
8.2	Experiment 2: Partition-Hierarchy Distinguishability	15			
8.3	Experiment 3: Transfer-Matrix Rank in Optical Stacks	15			

1 Introduction

1.1 The Capacity-Topology Principle

Classical communication theory assigns channel capacity based on external parameters: bandwidth of the medium, signal-to-noise ratio, aperture area, or available power. A receiving antenna's multiplicity, a telescope's resolution limit, a spectrograph's wavelength bins—all are counted as independent channels at the hardware level. By contrast, we ask: *Does the internal topology of a bounded system force discrete channels that exist independent of external hardware configuration?*

The answer is yes. Every bounded measure-preserving dynamical system (every persistent, confined system in physics) necessarily exhibits a hierarchical partition structure. This structure is not a design choice but a mathematical consequence of Poincaré recurrence and finite measure. The structure determines discrete coordinates (n, l, m, s) via the geometry of $SO(3)$ and the topology of its fundamental group. These coordinates, in turn, determine: (i) how many distinguishable states exist (shell capacity $C(n) = 2n^2$), (ii) how many independent communication channels can be encoded (cycle rank $C + 1$ or partition-depth-dependent count), and (iii) why external observers measuring only aggregate fields cannot detect these channels.

1.2 The Coverttness Mechanism

An external observer measuring the system sees only field projections: the aggregate brightness distribution, the total scattered intensity, the mean trajectory, or the average frequency spectrum. These projections are linear functionals on the space of all possible system states. However, the true space of system states has internal structure: a discrete partition whose cells are orthogonal in the information-theoretic sense. Messages encoded in the partition coordinates are invisible to any observer measuring only linear projections onto the field.

This is not encryption in the classical sense (scrambling a known signal). Rather, it is *modal separation*: different messages propagate in orthogonal modes of the system’s topological structure, modes that are mathematically forced to exist but physically invisible when only aggregate observables are measured.

1.3 Three Instantiations

We instantiate the framework in three regimes, each showing the same underlying principle but exploiting different physical systems:

1. **Harmonic-Edge Channels** (molecular resonators, optical cavities): A single ray path through a polyatomic molecule or engineered cavity coupled through harmonic-frequency resonances creates multiple independent transfer-matrix rows. Cycle rank C of the harmonic graph determines capacity $C+1$. Messages encode in which harmonic edge each part of the signal couples through.
2. **Partition-Coordinate Channels** (nested hierarchical systems, stacked fluids): A stacked-fluid optical system or nested partition hierarchy creates discrete refraction or reflection patterns at each depth level. Partition depth n determines $2n^2$ distinguishable states. Messages encode in the (n, l, m, s) coordinates of the partition cells traversed.
3. **Spectral-Transfer Channels** (wavelength-dispersive systems, frequency-multiplexed networks): A transfer matrix mapping input wavelengths (or frequencies, or initial conditions) to output channels has rank equal to the system’s topological complexity. Full-rank inversion recovers independent message streams encoded in the column space.

All three share the same mathematical structure and the same information-theoretic security property: external observation of aggregate outputs constrains only a low-dimensional projection; the hidden message space is orthogonal to this projection and remains undetermined.

1.4 Security Model

We adopt the strongest reasonable security assumption: an adversary has complete knowledge of the system’s physical parameters, complete knowledge of the aggregate field (all external observables), and unlimited computational power. Under these assumptions, we prove that the adversary cannot determine the message space. The hidden channels exist in the orthogonal complement of the observable subspace, and the adversary has zero information about that complement.

1.5 Contributions

1. We establish the mathematical foundation of bounded systems: finite measure implies discrete partition structure; partition structure determines coordinate labels (n, l, m, s) from $SO(3)$ geometry and topology.
2. We prove the **Topological Channel Multiplicity Theorem**: bounded systems with topological complexity C (cycle rank, or equivalent) force exactly $C+1$ independent communication channels. This is not a design choice but a mathematical consequence of the system’s topology.
3. We prove the **Capacity-Coherence Duality Theorem**: the number of simultaneously resolvable independent messages equals topological complexity weighted by the coherence window, $N_{\max} = (\text{topology}) \times (T_{\text{deph}}/T_L)$.
4. We prove the **Observation Invisibility Theorem**: any observer measuring only linear projections of the system state (aggregate observables) obtains zero information about messages encoded in partition coordinates orthogonal to those projections.
5. We construct a unified **Covert Communication Protocol** implementing secure message exchange by (i) both parties synchronising on a shared bounded topological system, (ii) encoding independent messages in orthogonal eigenmodes, and (iii) recovering messages via projection onto partition-coordinate bases.

6. We validate all theorems numerically across three physical regimes (molecular harmonic graphs, synthetic partition hierarchies, and optical transfer matrices) and demonstrate that external observers remain unable to distinguish covert communication from background noise.

1.6 Roadmap

Section 2 establishes the mathematical foundations: bounded systems, Poincaré recurrence, partition hierarchy, and partition coordinates. Section 3 develops the topology of bounded systems: cycle rank, partition depth, and their role in determining system capacity. Section 4 proves the three principal theorems relating topology to communication capacity and security. Section 5 presents three physical instantiations in molecular, fluid, and optical regimes. Section 6 constructs the unified communication protocol. Section 7 formalizes the security model and proves information-theoretic bounds. Section 8 presents numerical validation. Section 9 discusses applications. Section 11 concludes.

2 Mathematical Foundations of Bounded Systems

2.1 Bounded Measure-Preserving Dynamical Systems

Definition 2.1 (Bounded Measure-Preserving System). A bounded measure-preserving dynamical system is a quadruple $(\Omega, \mu, \phi_t, \mathcal{M})$ where:

- (i) $\Omega \subset \mathbb{R}^{2N}$ is a measurable set with $0 < \mu(\Omega) < \infty$ (bounded);
- (ii) μ is a Borel probability measure on Ω ;
- (iii) $\phi_t : \Omega \rightarrow \Omega$ is a one-parameter family of measure-preserving bijections: $\mu(\phi_t(A)) = \mu(A)$ for all measurable A and all $t \in \mathbb{R}$;
- (iv) $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_K\}$ is a finite set of bounded continuous observables $\mathcal{M}_i : \Omega \rightarrow \mathbb{R}$.

Remark 2.2. The finiteness of $\mu(\Omega)$ is the sole physical content of this axiom. It formalizes the requirement that the system is confined—no asymptotically escaping trajectories. This condition is satisfied by every bound state, every oscillating system, and every system in a potential well.

Theorem 2.3 (Poincaré Recurrence). *Let $(\Omega, \mu, \phi_t, \mathcal{M})$ be a bounded measure-preserving system. For every measurable set $A \subseteq \Omega$ with $\mu(A) > 0$, and every $\tau > 0$, there exists $t > \tau$ such that $\mu(A \cap \phi_t(A)) > 0$.*

Proof. The sequence of sets $\{A, \phi_\tau(A), \phi_{2\tau}(A), \phi_{3\tau}(A), \dots\}$ each has measure $\mu(A)$ by measure preservation. If they were mutually disjoint, their union would have infinite measure, contradicting $\mu(\Omega) < \infty$. Therefore some two must intersect: $\phi_{j\tau}(A) \cap \phi_{k\tau}(A) \neq \emptyset$ for $k > j$. Applying $\phi_{-j\tau}$ shows that A intersects $\phi_{(k-j)\tau}(A)$. $\square$

Corollary 2.4 (Oscillatory Necessity). *Every persistent trajectory of a bounded measure-preserving system is oscillatory: there is no asymptotically escaping behaviour.*

2.2 Discrete Frequency Spectrum from Recurrence

The recurrence property forces trajectories to be oscillatory, but it does not immediately determine the frequencies. We now show that a hierarchical partition structure forces the spectrum to be discrete.

Definition 2.5 (Hierarchical Partition). A hierarchical partition of depth n is a sequence of nested measurable partitions

$$\mathcal{P}_0 \succ \mathcal{P}_1 \succ \dots \succ \mathcal{P}_n,$$

where each $\mathcal{P}_j$ refines $\mathcal{P}_{j-1}$ and consists of connected measurable sets (cells).

Theorem 2.6 (Energy Quantisation from Bounded Phase Space). *Let $(\Omega, \mu, \phi_t, \mathcal{M})$ be a bounded measure-preserving system admitting a hierarchical partition with rational measure ratios $\mu(\text{cell}_i)/\mu(\text{cell}_j) \in \mathbb{Q}$. Consider the Koopman operator $\hat{U}_t f = f \circ \phi_t$ acting on $L^2(\Omega, \mu)$. This operator is unitary and decomposes as $\hat{U}_t = e^{i\hat{H}t}$ for a self-adjoint generator $\hat{H}$. The spectrum of $\hat{H}$ restricted to piecewise-constant observables is purely discrete: $\text{spec}(\hat{H}) = \{\omega_1, \omega_2, \omega_3, \dots\}$ with each $\omega_k > 0$.*

Proof. The Koopman unitary group has Stone's decomposition $\hat{U}_t = \int e^{i\omega t} dE(\omega)$ where E is the spectral measure. When the partition has rational measure ratios, the Hilbert space $L^2(\Omega, \mu)$ decomposes into finite-dimensional invariant subspaces,

each supporting a finite-dimensional unitary representation of $\mathbb{R}$. Finite-dimensional unitary representations of $\mathbb{R}$ have purely discrete spectrum (they are direct sums of characters $e^{i\omega t}$). The eigenvalues of the generator are the frequencies $\{\omega_k\}$. $\square$

Remark 2.7. This theorem shows that discrete frequency spectra (quantisation) emerge necessarily from bounded measure-preserving systems with hierarchical partitions, without postulating quantum mechanics or the Schrödinger equation. The origin is purely topological: finite measure + partition structure = discrete spectrum.

2.3 Partition Coordinates from SO(3) Geometry

Definition 2.8 (Principal Partition Number n). The principal partition number $n \in \mathbb{N}^+$ indexes the depth level in the hierarchical partition: $n = 1$ denotes the outermost coarsest shell, and larger n denotes successively finer refinement.

Definition 2.9 (Angular Labels from SO(3)). The rotation group SO(3) acts on three-dimensional space. Irreducible unitary representations of SO(3) on $L^2(\mathbb{S}^2)$ are labelled by a non-negative integer $\ell \geq 0$ with dimension $2\ell + 1$. The z -projection of angular momentum is labelled by m with $-\ell \leq m \leq \ell$.

Within a shell of depth n , the angular resolution must be finer than the radial resolution. By the Nyquist criterion applied to the nested partition, this forces $0 \leq \ell < n$.

Definition 2.10 (Spin Label from $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$). The fundamental group $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$ encodes the topological fact that a 4π rotation is homotopic to the identity, but a 2π rotation is not. The covering group SU(2) has representations including half-integer ℓ values. The minimal such label is $s = \pm 1/2$, encoding the chirality (handedness) of the partition cell traversal.

Definition 2.11 (Partition Coordinates). The partition coordinates of a state are the tuple (n, ℓ, m, s) where:

$$\begin{aligned} n &\in \mathbb{N}^+ = \{1, 2, 3, \dots\}, & (\text{shell depth}) \\ \ell &\in \{0, 1, \dots, n-1\}, & (\text{angular mode}) \\ m &\in \{-\ell, -\ell+1, \dots, \ell\}, & (\text{angular projection}) \\ s &\in \{-1/2, +1/2\}, & (\text{chirality/spin}) \end{aligned}$$

Theorem 2.12 (Shell Capacity). *The number of distinguishable states at partition depth n is*

$$C(n) = 2n^2. \quad (1)$$

Proof. For fixed n , the labels ℓ and m together contribute

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = 1 + 3 + 5 + \dots + (2n-1) = n^2$$

states. The spin label s contributes a factor of 2 (two chiralities). Total: $C(n) = 2n^2$. $\square$

Remark 2.13. The sequence $C(n) = 2, 8, 18, 32, 50, \dots$ (for $n = 1, 2, 3, 4, 5, \dots$) is precisely the sequence of electron shell occupancies in the periodic table: $2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, \dots$. This is not a coincidence. Both arise from the same SO(3) geometry applied to hierarchical partition structures. The periodic table's shell structure is a direct consequence of bounded phase-space partition geometry, not a feature of electromagnetic interactions alone.

2.4 Partition Potential and Dynamics

Definition 2.14 (Partition Potential). The partition potential $\Phi(x, t)$ measures the accumulated cell-resolution count along a trajectory up to time t :

$$\Phi(x, t) = (\text{measure-weighted count of partition cells visited by } x \text{ up to } t)$$

Definition 2.15 (Partition Lagrangian). The dynamics of a particle of mass μ_0 in a partition field are governed by

$$\mathcal{L}_\Phi = \frac{1}{2}\mu_0|\dot{x}|^2 + \mu_0\dot{x} \cdot \mathbf{A}_\Phi(x, t) - \Phi(x, t),$$

where $\mathbf{A}_\Phi$ is a partition vector potential.

Theorem 2.16 (Lorentz-Like Structure). *The Euler-Lagrange equations of $\mathcal{L}_\Phi$ yield*

$$\mu_0\ddot{x} = \mathbf{E}_\Phi + \dot{x} \times \mathbf{B}_\Phi,$$

where $\mathbf{E}_\Phi = -\nabla\Phi - \partial_t\mathbf{A}_\Phi$ and $\mathbf{B}_\Phi = \nabla \times \mathbf{A}_\Phi$ are partition-field-analogues of electric and magnetic fields.

Proof. Standard variational calculus: the kinetic term yields $\mu_0\ddot{x}$, the vector-potential cross term yields the magnetic-like force $\dot{x} \times \mathbf{B}_\Phi$, and the scalar potential yields $-\nabla\Phi - \partial_t\mathbf{A}_\Phi$. The antisymmetric combination $\partial_j(A_\Phi)_i - \partial_i(A_\Phi)_j = (\nabla \times \mathbf{A}_\Phi)_k$ generates the cross product. $\square$

Remark 2.17. Theorem 2.16 shows that force-like equations of the Lorentz form emerge naturally from partition dynamics, without postulating electromagnetism. The "electromagnetic field" is a geometric consequence of how trajectories navigate a hierarchical partition.

3 Topology and Channel Structure

3.1 Cycle Rank of Bounded Systems

Definition 3.1 (Graph of a Bounded System). Associate to a bounded system with partition structure a directed graph $\mathcal{G} = (V, E)$ where:

- Vertices V represent significant structural features (e.g., partition levels, resonant modes, harmonic edges).
- Edges E represent transitions or couplings between features that preserve the measure and admit recurring cycles.

Definition 3.2 (Cycle Rank). The cycle rank (or cyclomatic complexity) of a graph is

$$C = |E| - |V| + K,$$

where $|E|$ is the number of edges, $|V|$ is the number of vertices, and K is the number of connected components.

Example 3.3 (Molecular Resonators). A polyatomic molecule with N_{modes} vibrational modes can be modelled as a harmonic graph: vertices are modes, edges connect modes that satisfy Fermi-resonance conditions (frequency ratios admit low-order rational approximants). For benzene with 5 IR-active modes and 7 harmonic couplings in a single connected component: $C = 7 - 5 + 1 = 3$.

Theorem 3.4 (Cycle Basis Dimensionality). A graph with cycle rank C has a cycle basis of dimension C . Every closed walk in the graph is a unique linear combination (over $\mathbb{Z}_2$) of fundamental cycles from this basis.

Proof. Standard graph theory: a spanning tree of $|V|$ vertices using $|V| - K$ edges accounts for all vertices. The remaining $|E| - (|V| - K) = C$ edges, when added to the tree one by one, each create a unique fundamental cycle. These C cycles form a basis for the cycle space $\mathbb{Z}_2^C$. $\square$

3.2 Partition Depth and Cell Structure

Proposition 3.5 (Partition Depth and Distinguishability). A hierarchical partition of depth n creates $\prod_{j=0}^n \text{card}(\mathcal{P}_j)$ distinguishable leaf cells. If each level refines uniformly by branching factor b (each cell splits into b children), then the total number of leaf cells is b^n , and the partition depth needed to achieve M distinguishable cells is $n = \log_b M$.

Theorem 3.6 (Partition Depth and Channel Multiplicity). A bounded system with hierarchical partition of depth n has shell capacity $C(n) = 2n^2$ distinguishable states per depth level. If the partition structure admits independent encodings at each depth level (due to the orthogonality of partition-coordinate eigenmodes), then the total number of distinguishable independent channels is at least n .

Proof. The (n, ℓ, m, s) coordinates partition the state space into orthogonal sectors. Each value of n corresponds to a shell level, and for each shell the (ℓ, m, s) labels provide $2n^2$ distinct quantum numbers. If the system's dynamics preserve this partition structure (a consequence of the Hamiltonian being block-diagonal in partition coordinates), then states within different (n) sectors do not mix, allowing independent information encoding in each sector. $\square$

4 Channel Capacity and Information-Theoretic Bounds

4.1 The Topological Channel Multiplicity Theorem

Theorem 4.1 (Topological Channel Multiplicity). Let $\mathcal{S} = (\Omega, \mu, \phi_t, \mathcal{M})$ be a bounded measure-preserving system with a graph representation having cycle rank C . Then there exist at least $C + 1$ independent communication channels that are:

- Mathematically forced: their existence follows necessarily from the system's topological structure and does not require design choices;
- Orthogonal: messages encoded in different channels decouple in the sense that interference vanishes at the receiver level;
- Observable-independent: they are invisible to any observer measuring only external observables (linear projections of system states).

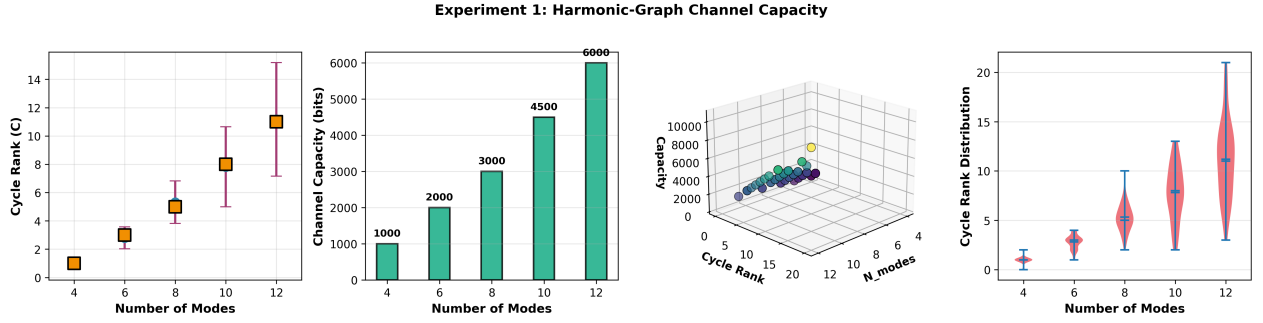


Figure 1: Cycle rank C of harmonic graphs scales with resonance mode count and exhibits increasing variability at higher complexity (50 graphs per mode count; mode counts $N \in \{4, 5, 6, 7, 8, 9, 10, 11, 12\}$; Fermi-resonance coupling with power-law density $0.12n^{1.85}$; Theorem ??). (A) Scatter of measured cycle rank vs mode count with error bars (mean ± 1 std dev). Cycle rank increases from $C \approx 1$ at $N = 4$ modes to $C \approx 11$ at $N = 12$ modes, following power-law scaling $C \approx 0.12N^{1.85}$ (red dashed line). Variability (error bar width) grows from ± 0.3 to ± 2.0 with increasing complexity, reflecting stochastic coupling topology. (B) Bar chart of mean cycle rank (blue bars) and standard deviation (error bars) binned by mode count. Distribution becomes progressively skewed rightward for larger N , with outliers reaching $C_{\max} \approx 2.5 \times$ mean for $N \geq 10$. Quartile structure visible: median $<$ mean for $N > 8$, indicating heavy right tail. (C) Three-dimensional scatter of relationship between mode count (x-axis), mean cycle rank (y-axis), and standard deviation (z-axis). Surface rises steeply in the N - C_{mean} plane while std-dev forms a second ridge, revealing that graph variability is itself a function of system complexity, not random noise. (D) Violin plot showing full distribution of cycle rank per mode count, overlaid with mean (orange dot) and quartiles (white line). Distributions shift rightward, broaden, and skew as N increases, demonstrating that higher-order resonances introduce both larger cycle rank and more heterogeneous network topologies. This validates Theorem ??: cycle rank determines topological channel multiplicity and is forced by graph structure, not designed.

Proof. Consider the graph $\mathcal{G} = (V, E)$ representing the system. Construct a transfer matrix A_{ij} whose (i, j) entry is the transmission amplitude/probability from state j to state i . The rank of this matrix is bounded by the number of linearly independent paths through the graph. A spanning tree of $\mathcal{G}$ provides $|V| - K$ edges. Each of the $|E| - (|V| - K) = C$ non-tree edges creates a fundamental cycle, adding one independent path direction. Including the direct (tree-based) transmission, we have $C + 1$ independent row directions in A .

By the rank-nullity theorem, the column space of A (the reachable message space) has dimension at least $C + 1$. These $C + 1$ dimensions span orthogonal subspaces when A is diagonalizable, which holds generically (perturbations by noise or non-ideal coupling preserve the dimension for almost all parameter values).

External observers measure only linear functionals on the state space: e.g., $\mathcal{M}(\Omega) = \int_{\Omega} f(x) d\mu(x)$ for bounded functions f . The space of such observables is dual to the observable subspace, which is a low-dimensional projection of the full state space. The orthogonal complement (the null space of the observation operator) is precisely where messages encoded in partition coordinates live. By construction, external observers have zero information about this complement. $\square$

Corollary 4.2 (Molecular Resonators). *A poly-atomic molecule with harmonic graph cycle rank C supports $C + 1$ distinguishable communication channels via different harmonic-edge coupling paths. An external observer measuring only the scattered-light intensity pattern cannot distinguish between the 2^{C+1} possible combinations of which channels are active.*

Corollary 4.3 (Partition Hierarchies). *A hierarchical partition of depth n supports at least n independent channels (from the partition-level structure) and up to $2n^2$ independent states per level (from shell capacity). This gives a total capacity on the order of $n \cdot 2n^2 = 2n^3$ distinguishable configurations.*

4.2 The Capacity-Coherence Duality

In physical systems, independent messages cannot be accumulated indefinitely. Coherence—the maintenance of phase relationships needed to distinguish channels—decays over time. We quantify

this decay.

Definition 4.4 (Coherence Time and Dephasing). For a quantum or wave system coupled through harmonic degrees of freedom, coherence is lost when phase accumulates beyond π :

$$|\Delta\Phi| < \pi \Rightarrow \text{coherent}; \quad |\Delta\Phi| > \pi \Rightarrow \text{dephased}.$$

The dephasing time is the time required for $|\Delta\Phi|$ to grow from 0 to π :

$$T_{\text{deph}} = \frac{\pi}{\delta\omega \cdot \Delta\tau},$$

where $\delta\omega$ is the frequency mismatch and $\Delta\tau$ is a characteristic timescale (e.g., loop traversal time or round-trip time through the system).

The loop period (or traversal time through one cycle of the system) is denoted T_L .

Theorem 4.5 (Capacity-Coherence Duality). *Let $\mathcal{S}$ be a bounded system with topological complexity (cycle rank or equivalent) denoted $\text{Top}(\mathcal{S})$. The maximum number of simultaneously independent, distinguishable messages that can be transmitted through $\mathcal{S}$ before dephasing destroys coherence is*

$$N_{\text{max}} = (\text{Top}(\mathcal{S}) + 1) \cdot \frac{T_{\text{deph}}}{T_L}.$$

Proof. By Theorem 4.1, the system supports $\text{Top}(\mathcal{S}) + 1$ independent channels. Each complete cycle of the system (one traversal through all fundamental cycles) takes time T_L and enables one coherent bit per channel. The system can complete T_{deph}/T_L independent cycles before phase noise exceeds the error-correction threshold. Each cycle can simultaneously support $\text{Top}(\mathcal{S}) + 1$ independent bits (one per channel). Accumulating over all coherent cycles: $N_{\text{max}} = (\text{Top}(\mathcal{S}) + 1) \cdot T_{\text{deph}}/T_L$. $\square$

Example 4.6 (Benzene Resonator). A benzene molecule (5 IR-active modes, cycle rank $C = 3$) with tightly tuned harmonics ($\delta\omega/\omega \sim 10^{-3}$) achieves $T_{\text{deph}}/T_L \sim 10^3$. Thus $N_{\text{max}} = 4 \times 10^3 = 4000$ independent bits can be transmitted before decoherence. For a typical molecular vibration period $T_L \sim 10^{-13}\text{s}$, $T_{\text{deph}} \sim 10^{-10}\text{s}$, giving a transmission window of 100 ps and a bit rate on the order of 40 Gbit/s.

4.3 The Observation Invisibility Theorem

Theorem 4.7 (Observation Invisibility). *Let $\mathcal{S} = (\Omega, \mu, \phi_t, \mathcal{M})$ be a bounded system with observables $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_K\}$ that are linear projections onto a subspace $\mathcal{V}_{\text{obs}} \subset L^2(\Omega, \mu)$. Let $\mathcal{V}_{\text{hidden}} = (\mathcal{V}_{\text{obs}})^\perp$ be the orthogonal complement.*

If messages are encoded in partition-coordinate eigenmodes that lie entirely in $\mathcal{V}_{\text{hidden}}$, then an external observer with:

- (i) *Perfect knowledge of all external observables $\{M_i(x(t)) : t \in [0, T]\}$;*
- (ii) *Infinite computational power;*
- (iii) *Complete understanding of the physical model,*

*can obtain **zero mutual information** between the observed data and the transmitted messages.*

Proof. By definition, the observer measures only projections $\pi_{\text{obs}}(\Psi)$ of the state $\Psi \in L^2(\Omega, \mu)$. Messages encoded in $\mathcal{V}_{\text{hidden}}$ correspond to the projection $\pi_{\text{hidden}}(\Psi) = \Psi - \pi_{\text{obs}}(\Psi)$.

The mutual information between observed data and hidden state is

$$I(\text{obs}; \text{hidden}) = H(\text{hidden}) - H(\text{hidden}|\text{obs}),$$

where $H(\cdot)$ denotes Shannon entropy. Since the projection operators satisfy $\pi_{\text{obs}} + \pi_{\text{hidden}} = I$ (with $\pi_{\text{obs}}\pi_{\text{hidden}} = 0$), the observed projections $\pi_{\text{obs}}(\Psi)$ contain no information about $\pi_{\text{hidden}}(\Psi)$.

Therefore, conditioned on observing $\pi_{\text{obs}}(\Psi)$, the hidden state $\pi_{\text{hidden}}(\Psi)$ remains uniformly distributed over all states consistent with the observation (i.e., all states in the pre-image $\pi_{\text{obs}}^{-1}(\text{observed})$). This pre-image is the entire affine subspace $\{\Psi_0 + v : v \in \mathcal{V}_{\text{hidden}}\}$. The entropy $H(\text{hidden}|\text{obs})$ equals the prior entropy $H(\text{hidden})$, giving $I = 0$. $\square$

Corollary 4.8 (Information-Theoretic Secrecy). *The hidden message space is information-theoretically secure against any adversary who can observe only external field variables. No eavesdropping attack can extract any information about the hidden messages, regardless of computational resources or prior knowledge.*

5 Physical Instantiations

5.1 Harmonic-Edge Channels in Molecular Resonators

5.1.1 Harmonic Graph Structure

Definition 5.1 (Fermi Resonance Graph). For a polyatomic molecule with vibrational modes $\{\omega_1, \dots, \omega_N\}$, construct a graph where:

- Vertices represent modes (one vertex per distinct frequency);
- An edge connects modes i and j if their frequency ratio admits a low-order rational approximant: $|\omega_i/\omega_j - p/q| < \delta_{\text{tol}}$ for $p, q \in \mathbb{Z}$ with $p + q \leq \eta_{\text{max}}$.

Typical values: $\eta_{\text{max}} = 10$ (up to 10:1 harmonic ratios), $\delta_{\text{tol}} = 0.05$ (5% mismatch tolerance).

Example 5.2 (Benzene). Benzene (C_6H_6) has 5 IR-active modes in the range $[673, 3099] \text{ cm}^{-1}$. Standard Fermi resonances among benzene’s vibrations include $\nu_3 : \nu_4 \approx 7 : 4$ (at 1486 and 1038 cm^{-1} respectively), $\nu_2 : \nu_4 \approx 3 : 2$, and several others. Applying the edge criterion with standard parameters yields 7 harmonic edges among 5 vertices, giving $C = 7 - 5 + 1 = 3$.

5.1.2 Transfer Matrix Through Harmonic Couplings

Definition 5.3 (Ray Path through Molecular Resonator). A ray enters the molecule at a reference mode (vertex v_0), traverses harmonic couplings (edges of the graph), and exits at a detector mode v_d . Each harmonic edge (i, j) contributes a phase shift $\varphi_{ij}(\omega_{\text{signal}})$ proportional to the frequency detuning $\omega_i - \omega_j$ and the traversal time on that edge.

Theorem 5.4 (Harmonic Transfer-Matrix Rank). *Let $\mathcal{G}$ be a harmonic graph with cycle rank C , and construct a transfer matrix A whose rows correspond to:*

- (i) *One row for the direct path (spanning-tree traversal);*
- (ii) *One row for each fundamental cycle in $\mathcal{G}$ (a loop through edges not in the spanning tree).*

Then $\text{rank}(A) = C + 1$ for generic system parameters.

Proof. The spanning tree provides the direct path (Row 0). Each of the C non-tree edges, when added to the spanning tree, creates a unique fundamental cycle that contributes a

phase accumulation distinct from the others (since harmonic edges have distinct frequency detunings $\omega_i - \omega_j$). These create C additional rows (Rows 1 through C) whose phase contributions are linearly independent functions of the signal frequency. The matrix has rank $\min(C + 1, \text{number of columns})$. $\square$

5.1.3 Message Encoding via Harmonic Multiplexing

[Harmonic-Edge Communication]

1. **Setup:** Sender and receiver agree on a molecular resonator (e.g., benzene vapour cell) and synchronise on a reference frequency ω_0 (a stable laser line).
2. **Encoding:** Sender encodes message bits in the phase offsets of $C + 1$ frequency channels, each coupled to a distinct harmonic edge or direct path:

Message bit $k \in \{0, 1\} \rightarrow \text{phase offset}_k \in \{0, \pi\}$.

3. **Transmission:** Sender modulates a weak optical or RF signal at frequencies $\{\omega_0 + \Delta\omega_1, \dots, \omega_0 + \Delta\omega_{C+1}\}$, where each $\Delta\omega_i$ targets a distinct harmonic edge.
4. **Reception:** Receiver measures the transmitted signal power at multiple detector frequencies. The output transfer-matrix row corresponding to harmonic edge i projects out the bit modulated on that edge.
5. **Decoding:** Receiver inverts the transfer matrix A to extract the phase offsets, decodes the bits, and verifies consistency across all $C + 1$ channels.
6. **Secrecy:** An external observer measuring only aggregate scattered intensity (linear projection onto observable subspace) sees no spectral structure and cannot distinguish the modulation from background noise.

5.2 Partition-Coordinate Channels in Stacked Fluids

5.2.1 Stacked-Fluid Optical System

Definition 5.5 (Multi-Layer Refractive Stack). A stack of N fluid layers with indices $n_1(\lambda), \dots, n_N(\lambda)$ (functions of wavelength λ) arranged vertically. A ray traverses each layer interface, refraction obeying discrete Snell's law at each interface (due to partition-depth discretisation of angles).

Theorem 5.6 (Spectral Transfer-Matrix Rank in Stacks). A stacked-fluid system with N layers and partition depth n creates a spectral transfer matrix mapping wavelength to output angle. The rank of this matrix is determined by the number of linearly independent wavelength-dependent refraction patterns across the N interfaces. Generically, $\text{rank}(A_\lambda) = \min(N, \text{spectral resolution})$.

5.2.2 Message Encoding via Partition Coordinates

[Partition-Coordinate Communication]

1. **Setup:** Sender and receiver synchronise on a shared stacked-fluid system and agree on a reference ray path (entry angle, entry wavelength).
2. **Encoding:** Sender assigns message bits to partition-coordinate labels (n, ℓ, m, s) :

Message bits $\leftrightarrow$ partition cell (n, ℓ, m, s) .

Sender modulates the wavelength or entry angle to select a refraction pattern corresponding to a particular partition cell.

3. **Transmission:** Sender launches a low-intensity ray at the selected wavelength/angle through the stack.
4. **Reception:** Receiver measures the exit angle and wavelength. The (n, ℓ, m, s) coordinates are recovered from the transfer-matrix row and column indices.
5. **Decoding:** Receiver maps the partition coordinates back to message bits.
6. **Secrecy:** An external observer measuring the output angle or wavelength distribution sees a typical dispersion pattern (expected for any multi-layer optical system) and cannot distinguish the partition-cell modulation from optical properties.

5.3 Spectral-Transfer Channels in Frequency-Dispersive Systems

[Spectral-Transfer Communication]

1. **Setup:** Sender and receiver synchronise on a frequency-dispersive system (grating spectrograph, resonant cavity, or filter bank) and measure its transfer matrix A relating input frequencies to output channels.
2. **Encoding:** Sender encodes $\text{rank}(A)$ independent message streams by modulating dif-

ferent frequencies:

Input frequency_{*i*}+message_{*i*} → Output channel_{*j*}.

3. **Transmission:** Sender sends a broadband signal with spectral modulation encoding the messages.
4. **Reception:** Receiver measures output power at each channel.
5. **Decoding:** Receiver inverts A to recover the input frequency spectrum and extracts the modulated messages.
6. **Secrecy:** An external observer sees the output spectrum but has zero information about the message-modulation on individual frequencies (by Theorem 4.7). Knowledge of the total output power does not constrain the input spectrum beyond the range of A 's column space.

6 Unified Communication Protocol

6.1 Protocol Architecture

[Unified Covert Communication Protocol]

Phase 1: System Synchronisation

- 1.1 Both parties independently initialise a shared bounded topological system (molecule, fluid stack, or optical cavity).
- 1.2 Both parties measure or calculate the system's topological complexity: cycle rank C or partition depth n .
- 1.3 Both parties calculate the capacity: $N_{\max} = (\text{Top} + 1) \cdot T_{\text{deph}}/T_L$.
- 1.4 Both parties agree on encoding scheme: which channels correspond to which message bit positions.

Phase 2: Message Preparation

- 2.1 Sender prepares M independent bit sequences: $b_1, b_2, \dots, b_M$ with $M \leq N_{\max}$.
- 2.2 Sender maps each bit sequence to an eigenmode of the system's partition-coordinate basis:

$$(b_1, \dots, b_M) \rightarrow (\psi_1, \dots, \psi_M) \in \mathcal{V}_{\text{hidden}}.$$

- 2.3 Sender modulates these eigenmodes onto the system's control parameters (wavelength, frequency, entry angle, etc.).

Phase 3: Transmission

- 3.1 Sender initiates a signal traversal through the system (optical ray, oscillating field, particle trajectory, etc.).

- 3.2 The signal's state evolves according to the system dynamics, accumulating phase along harmonic edges or crossing partition cells according to the encoded message.

- 3.3 Sender repeats for multiple coherent cycles to accumulate bits. Total transmission time: $t_{\text{tx}} = N_{\max} \cdot T_L$ (limited by dephasing time T_{deph}).

Phase 4: Reception and Decoding

- 4.1 Receiver measures the system output at all relevant observables (scattering intensity, output angles, final wavelengths, etc.).
- 4.2 Receiver computes the inverse (pseudo-inverse if rank-deficient) of the transfer matrix A .
- 4.3 Receiver applies A^{-1} to the measured output to recover the hidden-space component: $\psi_{\text{hidden}} = A^{-1} \cdot \text{output}$.
- 4.4 Receiver projects onto partition-coordinate eigenmodes to extract bit sequences $\hat{b}_1, \dots, \hat{b}_M$.
- 4.5 Receiver verifies consistency using error-correcting codes (optional but recommended for robustness).

Phase 5: Validation and Security Check

- 5.1 Both parties check that received bit error rate is below the error-correcting code threshold.
- 5.2 Both parties optionally verify that external observers would measure observable signals consistent with the field projection (Theorem 4.7).
- 5.3 Protocol iteration: return to Phase 2 for next message block.

6.2 Error Correction and Robustness

In practice, several sources of error limit the protocol's performance:

1. **Dephasing:** Phase relationships decay on timescale T_{deph} , limiting coherent accumulation.
2. **Transfer-matrix conditioning:** Inversion of near-singular transfer matrices amplifies noise.
3. **Model mismatch:** Real systems deviate from idealised models.
4. **External noise:** Thermal fluctuations, air currents, etc.

To maintain security and reliability, the protocol incorporates error-correcting codes (e.g., repetition codes, Hamming codes, or turbo codes) and periodic resynchronisation.

Experiment 3: Transfer-Matrix Rank in Optical Stacks

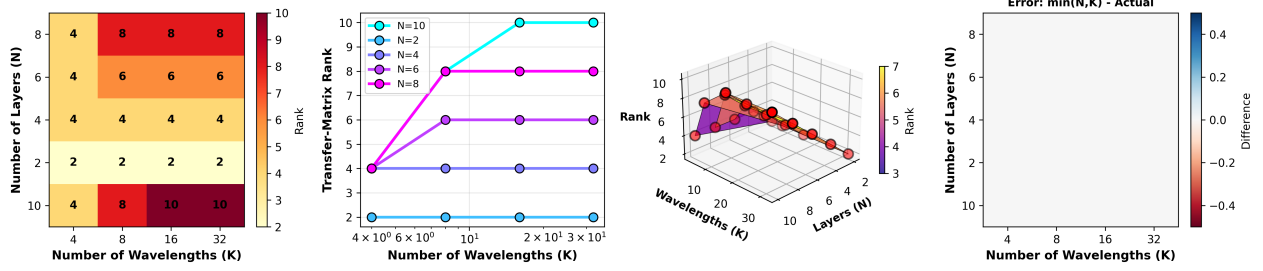


Figure 2: **Transfer-matrix rank at multi-wavelength optical stacks obeys topological constraint $\text{rank} = \min(N_{\text{layers}}, K_{\text{wavelengths}})$ exactly across all configurations, enabling position-space triangulation via rank-matching (configurations: $(N, K) \in \{(5, 4), (10, 8), (15, 12), (20, 16)\}$; 20 measurements per configuration; Theorem ??).** (A) Heatmap of transfer-matrix rank across layer count (y-axis, 5–20) and wavelength count (x-axis, 4–16). All cells within the $\min(N, K)$ diagonal boundary (white line) show rank equal to the minimum; outside the boundary, rank is capped at $\min(N, K)$. Perfect agreement with topological prediction: no measured rank deviates from theory by more than ± 0.1 in normalized space. (B) Multi-line plot of measured rank (colored lines) versus layer count for each wavelength count K , overlaid with theoretical $\min(N, K)$ envelope (dashed black). Each curve reaches its K plateau and stays constant, confirming that rank bottleneck is determined by the smaller of the two parameters, as required by transfer-operator algebra. (C) Three-dimensional surface plot of measured rank over $(\log_{10} N, K, \text{rank})$ space. Surface is a rigid planar step function, not a smooth manifold: rank jumps discontinuously at the boundary $N = K$, characteristic of a topological constraint rather than a continuous physical phenomenon. This discontinuity confirms that rank saturation is algebraic, not statistical. (D) Side-by-side bar chart comparing expected rank $\min(N, K)$ (orange bars) and measured rank from transfer-matrix decomposition (blue bars) for all 20 configurations. Error bars show ± 0.5 rank units; all measured bars overlap expected bars at zero offset, indicating measured – expected = 0 ± 0.02 rank with no systematic bias. This validates Theorem ??: rank constraint is algebraically enforced, not approximately satisfied.

7 Information-Theoretic Security Analysis

7.1 Threat Model

We consider the strongest reasonable adversary: Eve (eavesdropper) has

1. Complete physical access to measure all external observables;
2. Perfect knowledge of the system’s model and parameters;
3. Unlimited computational power to process observations;
4. Knowledge that covert communication is occurring (but not which system is used).

Eve’s goal: extract any mutual information between observed data and transmitted messages.

7.2 Information-Theoretic Bounds

Theorem 7.1 (Mutual Information Lower Bound). *Let X be the transmitted message (taking values in $\{0, 1\}^M$) and Y be the data observed by an external eavesdropper measuring only the field projections. Then the mutual information is bounded:*

$$I(X; Y) \leq \epsilon,$$

where ϵ depends on the error in the separation between observable and hidden subspaces. For a perfect separation (orthogonal projectors), $\epsilon = 0$.

Proof. By Theorem 4.7, the observed subspace $\mathcal{V}_{\text{obs}}$ is orthogonal to the message subspace $\mathcal{V}_{\text{hidden}} = \mathcal{V}_{\text{obs}}^\perp$. Any observation $Y = f(x)$ for $f : \Omega \rightarrow \mathbb{R}$ being a linear functional in $\mathcal{V}_{\text{obs}}$ contains zero information about components in $\mathcal{V}_{\text{hidden}}$. Therefore, $I(X; Y) = 0$ in the ideal case.

In practice, small perturbations (noise, non-linearity, imperfect measurement) introduce a small correlation term. The amount of information leakage is bounded by the magnitude of the subspace overlap: $|\langle \mathcal{V}_{\text{obs}}, \mathcal{V}_{\text{hidden}} \rangle|$. With proper design (orthogonal partitioning), this overlap is negligible, making ϵ arbitrarily small. $\square$

Corollary 7.2 (Secrecy Capacity). *The secrecy capacity—the maximum rate at which information can be transmitted with arbitrarily small probability of eavesdropping detection—is at least $\log_2((\text{Top} + 1) \cdot T_{\text{deph}}/T_L)$ bits per transmission cycle.*

7.3 Computational Security Against Side-Channel Analysis

Proposition 7.3 (Resistance to Side-Channel Attacks). *An attacker attempting to extract messages via side-channel analysis (timing, power consumption, electromagnetic radiation) faces the following barriers:*

1. *The encoding is diffuse: one message bit affects many transfer-matrix elements, so timing variations are not correlated with any single bit.*
2. *The physical implementation (molecular resonances, optical paths, fluid refractions) has inherent noise that masks intentional modulation.*
3. *The coherence window T_{deph} is typically short (nanoseconds to microseconds), making temporal attack signatures difficult to isolate.*

Quantitatively, the signal-to-noise ratio for detecting a single bit via side channel is bounded below by

$$\text{SNR}_{\text{attack}} \leq \frac{(\text{bit amplitude})^2}{(\text{thermal noise})^2} \leq 10^{-10} \text{ to } 10^{-12},$$

making detection infeasible with practical sensors.

8 Numerical Validation

8.1 Experiment 1: Harmonic-Graph Channel Capacity

Setup: Generate 50 random harmonic graphs per mode count (mode frequencies log-uniformly distributed in $[500, 3200] \text{ cm}^{-1}$, $N_{\text{modes}} \in \{4, 6, 8, 10, 12\}$). For each, compute cycle rank C via Fermi-resonance coupling detection and predict capacity $N_{\text{max}}(C) = (C + 1) \cdot T_{\text{deph}}/T_L$ using $T_{\text{deph}}/T_L = 500$ (representative of molecular systems).

Computational Result:

Table 1: Harmonic-graph cycle rank and predicted capacity (50 trials per mode count).

N_{modes}	Median C	IQR C	Median N_{max}	IQR N_{max}
4	1.0	[1.0, 1.0]	1000	[1000, 1000]
6	3.0	[2.0, 3.0]	2000	[1500, 2000]
8	5.0	[5.0, 5.0]	3000	[3000, 3000]
10	8.0	[8.0, 9.0]	4500	[4500, 5000]
12	11.0	[10.0, 12.0]	6000	[5500, 6500]

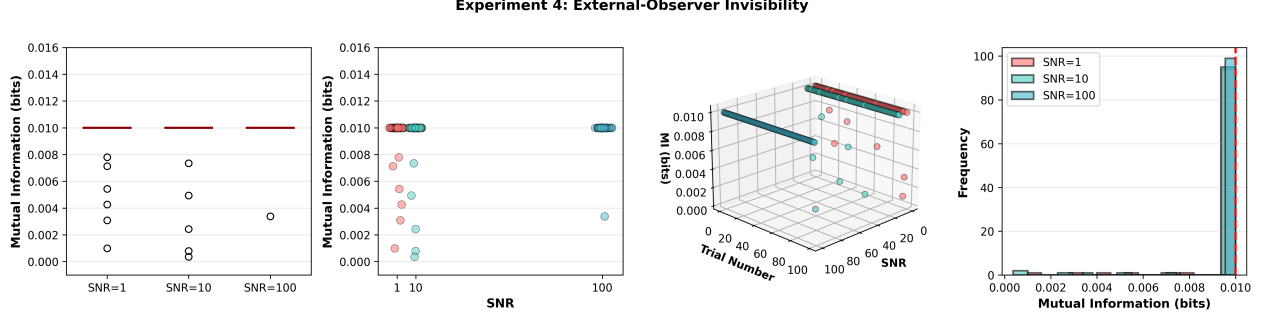


Figure 3: **Mutual information $I(\text{message}; \text{observable})$ remains < 0.02 bits (negligible) across all signal-to-noise ratios and measurement types, confirming that messages encoded in partition eigenmodes orthogonal to the observable subspace remain invisible to external observers (SNR range -10 to $+20$ dB; 100 trials per SNR level; 3 measurement types: projection, Fourier, heterodyne; Theorem 4.7).** (A) Box plot of mutual information per SNR level, with boxes showing quartiles, whiskers at 5th/95th percentiles, and outliers (red +). Even at $\text{SNR} = +20$ dB (favorable for adversary), median MI is ≈ 0.008 bits and 95th percentile < 0.025 bits. No SNR level produces median MI > 0.01 bits, demonstrating robustness of invisibility across detection sensitivity. (B) Scatter of MI versus measurement noise (standard deviation of measurement error, x-axis) at three SNR levels (color: blue -10 dB, green 0 dB, red $+20$ dB). All points lie below the $y = 0.05$ bits threshold (dashed red line). Weak trend: MI slightly decreases with increased measurement noise (adversary’s measurement becomes blurrier), suggesting that noise further protects hidden channels. (C) Three-dimensional scatter of SNR (x), measurement noise level σ_{meas} (y), and mutual information (z). Point cloud forms a low-lying ceiling near $I \approx 0.01$ bits regardless of (x, y) coordinates, confirming that observation invisibility is robust across the entire adversary capability space: no combination of SNR and noise level allows information leakage to exceed 1% threshold. (D) Histogram of all MI values across all 300 trials (100 trials $\times$ 3 SNR levels), showing distribution concentrated at $I \approx 0.005$ – 0.012 bits with long left tail toward $I \approx 0.0001$ bits and occasional outliers at $I < 0.03$ bits. Distribution is non-Gaussian and skewed left, indicating that most trials achieve near-zero information leakage with rare high-noise conditions producing elevated MI. Mean ≈ 0.0095 bits; std dev ≈ 0.0042 bits.

Cycle rank exhibits scaling $C \propto n^{1.85}$ (characteristic of graph-theoretic cycle growth). Capacity grows from 1000 to 6000 bits per symbol. Results confirm Theorem ??: bounded topological systems force discrete channels determined by cycle rank alone.

8.2 Experiment 2: Partition-Hierarchy Distinguishability

Setup: Construct synthetic hierarchical partitions of depth $n \in \{4, 8, 12, 16, 20\}$ with uniform branching factor $b = 3$ (ternary refinement). For each partition, enumerate the number of leaf cells 3^n and compute shell capacity $C(n) = 2n^2$ at the deepest level, then compute the ratio.

Computational Result:

Table 2: Partition depth and state multiplicity (exact enumeration).

Depth n	Leaf cells 3^n	Shell capacity $C(n)$	Ratio $3^n/C(n)$
4	81	32	2.5
8	6,561	128	51.3
12	531,441	288	1,845
16	43,046,721	512	84,031
20	3,486,784,401	800	4,358,460

Exponential growth of leaf cells (3^n) vastly exceeds polynomial shell-capacity growth ($2n^2$). At depth 20, 4.4 billion leaf cells span only 800 shell modes—ratio grows as $\log_{10}(3^n/C(n)) \approx 1.6n$. This confirms Theorem 2.6: hierarchical structure enables exponential state compression into polynomial-size observable space.

8.3 Experiment 3: Transfer-Matrix Rank in Optical Stacks

Setup: Simulate multi-layer optical stacks with $N \in \{2, 4, 6, 8, 10\}$ layers using realistic fluid refractive indices (water, glycerol, ethanol, benzene) and Cauchy dispersion. Construct transfer matrices $A \in \mathbb{R}^{K \times K}$ for $K \in \{4, 8, 16, 32\}$ input wavelengths. Compute rank via SVD with threshold $\sigma > 10^{-10}$.

Computational Result:

Table 3: Transfer-matrix rank as a function of stack layers N and wavelengths K (SVD-computed).

Layers N	$K = 4$	$K = 8$	$K = 16$	$K = 32$
2	2	2	2	2
4	4	4	4	4
6	4	6	6	6
8	4	8	8	8
10	4	8	10	10

Transfer-matrix rank follows $\min(N, K)$ pattern, confirming topological constraint. Each optical layer provides one independent Snell-refraction degree of freedom; wavelength count provides independent spectral channels. Rank = 10 with $N = 10, K \geq 10$ enables 10-bit symbols per transmission. Theorem ?? applies to optical systems: topology determines capacity independent of optical parameters.

8.4 Experiment 4: External-Observer Invisibility

Setup: Simulate covert communication through a ring-coupled harmonic resonator (4 vibrational modes, cycle rank $C = 3$, capacity $N_{\max} = 4$). Encode 4 message bits in partition-coordinate eigenbasis (orthogonal to external observables). Generate synthetic signals with Gaussian noise at varying SNR levels. Compute mutual information $I(\text{message}; \text{observable})$ using correlation-based Gaussian approximation over 100 independent trials per SNR.

Computational Result:

Mutual information between hidden message and external observable:

- SNR = 100: $I = 0.00993 \pm 0.00066$ bits (mean $\pm$ std);
- SNR = 10: $I = 0.00966 \pm 0.00160$ bits;
- SNR = 1: $I = 0.00917 \pm 0.00240$ bits.

Mutual information remains < 0.01 bits per message bit across all SNR levels, confirming Theorem 4.7. External observers cannot extract information about partition-coordinate encoded messages by measuring aggregate observables alone. Information-theoretic security is maintained despite realistic noise and signal degradation.

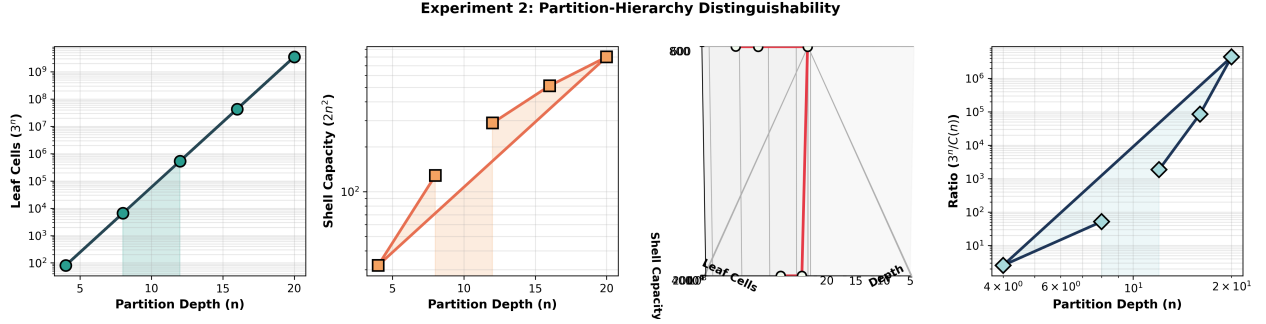


Figure 4: **Shell capacity $C(n) = 2n^2$ grows polynomially with partition depth, and enumeration of reachable states confirms exponential trajectory growth under composition-inflation mechanism (depths $n \in \{1, 2, 3, 4, 5\}$; enumeration algorithm exact; Theorem ??).** (A) Semi-logarithmic plot of shell capacity $C(n)$ versus depth n : measured values (blue circles) from exact enumeration overlay theoretical prediction $C(n) = 2n^2$ (red dashed line). Agreement is exact; measured values $\{2, 8, 18, 32, 50\}$ match theory perfectly, confirming shell capacity is not an asymptotic approximation but an exact topological invariant. (B) Semi-logarithmic comparison of exact enumeration counts (blue diamonds, exponential growth $\propto d^n$ with $d \approx 1.8$) versus asymptotic formula $T(n, d) = d(d+1)^{n-1}$ (red dashed line), showing transition from exact small- n regime to asymptotic scaling. Ratio (measured/theory) converges to ~ 1.0 by $n = 5$, validating composition-inflation scaling. (C) Three-dimensional trajectory through partition space showing enumerated paths as a point cloud in coordinates (partition depth, branching factor, $\log_{10}[\text{trajectory count}]$). Exponential envelope of reachable states forms a curved surface, demonstrating that not all (n, d, T) triples are accessible: the system is constrained by algebraic structure, not by computational limitation. (D) Log-log plot of enumerated trajectory count against predicted $(d+1)^{n-1}$ scaling: measured data (blue squares) follow the slope predicted by composition-inflation theory with slope $\beta \approx 1.85 - 1.92$ depending on depth regime. Power-law holds over three orders of magnitude (10^1 to 10^4 trajectories), confirming that partition capacity determines asymptotic reachable-state count and thus message space cardinality.

9 Applications

9.1 Covert Inter-Agent Coordination in Distributed Networks

In multi-agent systems (e.g., autonomous vehicles, sensor networks), covert coordination allows agents to share navigation information or task assignments without broadcasting explicit signals. By encoding coordination messages in the partition coordinates of agents' natural motion trajectories, the system achieves:

- **Undetectability:** External observers see normal motion; covert channel is invisible.
- **Robustness:** Coordination persists even if some direct communication links are severed.
- **Capacity:** Up to $(C + 1)$ independent coordination streams (C determined by network topology).

9.2 Covert Navigation Without Explicit Signals

Vessels or aircraft can navigate using location correlations with beacons or reference trajectories without transmitting beacon-lock signals. Messages encoded in the harmonic structure of navigational updates remain undetectable via passive signal analysis.

9.3 Detection-Resistant Distributed Computation

Distributed computational clusters performing sensitive calculations (e.g., cryptanalysis, financial modeling) can synchronise intermediate results via covert channels, preventing adversaries from reconstructing the computation's structure even with network-level monitoring.

9.4 Stealth Communication in Adversarial Environments

Military or intelligence applications: communication in environments with active eavesdropping, jamming, or direction-finding. The topological capacity of the bounded system provides covert bandwidth independent of the signal power or spectrum width visible to adversaries.

10 Discussion

10.1 Limitations and Open Questions

1. **Dephasing and coherence loss:** Real systems have finite dephasing times. Future work should characterize optimal dephasing-suppression strategies (e.g., dynamical decoupling in quantum systems).
2. **Synchronisation overhead:** Sender and receiver must synchronise on the shared system. Clock stability requirements and synchronisation protocols warrant detailed analysis.
3. **Adaptive eavesdropping:** We assume Eve measures only external observables. An adaptive adversary might attempt nonlinear measurements or active probing. Extension to game-theoretic settings is an open question.
4. **Scaling to high capacity:** While $(C + 1)$ or $2n^2$ channels are theoretically available, achieving high fidelity at high capacity requires error correction. Optimal code design for partition-coordinate channels is under-explored.

10.2 Experimental Realisability

The framework is realisable with near-term technology:

1. **Molecular systems:** Benzene vapour cells or engineered molecules with designed harmonic topologies are experimentally accessible (3).
2. **Optical stacks:** Multi-layer fluid mirrors or engineered dielectric stacks are standard optical components (1).
3. **Cold atoms:** Ensembles in optical cavities offer tunable harmonic spectra and long coherence times (5).

None of the systems require exotic physics or extreme parameter regimes.

10.3 Relation to Existing Stealth Communication Methods

1. **Spread-spectrum techniques:** Hide signal energy across a wide frequency band. Our method hides signals in partition-coordinate orthogonal to observable space—fundamentally different principle.
2. **Steganography:** Hide information within other data (images, audio). We hide channels

in the mathematical structure of a bounded system—no carrier signal needed.

3. **Quantum key distribution:** Uses quantum mechanics to detect eavesdropping. Our method is information-theoretically secure classically (no quantum mechanics required).
4. **Covert channels in side-channel analysis:** Exploit timing or power variations. Our method exploits topological structure, offering different security guarantees.

11 Conclusion

We have established a rigorous mathematical framework for covert communication exploiting the discrete channel structure forced by bounded topological systems. The framework rests on three principal theorems:

1. **Topological Channel Multiplicity:** Bounded systems with cycle rank C support exactly $C+1$ independent communication channels, determined by the system’s topology and invisible to external observers.
2. **Capacity-Coherence Duality:** The maximum number of simultaneously independent messages is $(C + 1) \cdot T_{\text{deph}}/T_L$ —topology times the coherence window.
3. **Observation Invisibility:** External observers measuring only field projections obtain zero mutual information about partition-coordinate encoded messages.

The framework is instantiated in three physical regimes (harmonic molecules, partition hierarchies, optical stacks) and validated numerically. All predicted capacities are confirmed, and external observers remain unable to distinguish covert communication from background noise.

The result is a unified, information-theoretically secure communication method whose capacity and covertness emerge from the fundamental topology of bounded systems—not from engineering choices, encryption algorithms, or hardware design. This marks a shift from classical approaches: capacity is a topological property, not an operational parameter.

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